BUILD AND EVALUATE A LINEAR REGRESSION MODEL

House Price Prediction using California Housing Dataset

ABSTRACT

This project focuses on building and evaluating a Linear Regression model for predicting house prices using the California Housing Dataset. The workflow includes data loading, exploratory data analysis (EDA), preprocessing, model training, prediction, and evaluation. Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn were used for implementation. The model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. The results showed that Linear Regression can predict house prices with reasonable accuracy and helps in understanding relationships between housing features and house values.

1. INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions automatically. Regression is a supervised learning technique used for predicting continuous numerical values.

Linear Regression is one of the most commonly used regression algorithms. It establishes a relationship between independent variables and a dependent variable using a best-fit line.

In this project, a Linear Regression model was developed to predict house prices using the California Housing Dataset. The dataset contains information related to housing districts such as median income, house age, average rooms, population, occupancy, and geographical coordinates. The project demonstrates the complete machine learning workflow including data analysis, preprocessing, training, prediction, and evaluation.

2. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed to understand the structure and characteristics of the dataset.

The following analyses were carried out:

- Checking missing values
- Statistical summary of the dataset
- Correlation analysis
- Distribution visualization

Observations

- The dataset contained no missing values.
- Median income showed strong positive correlation with house prices.
- Latitude and longitude also influenced house values.

- Distribution plots helped in understanding the spread of house prices.

A correlation heatmap and distribution plots were generated using Matplotlib and Seaborn for better visualization of the dataset.

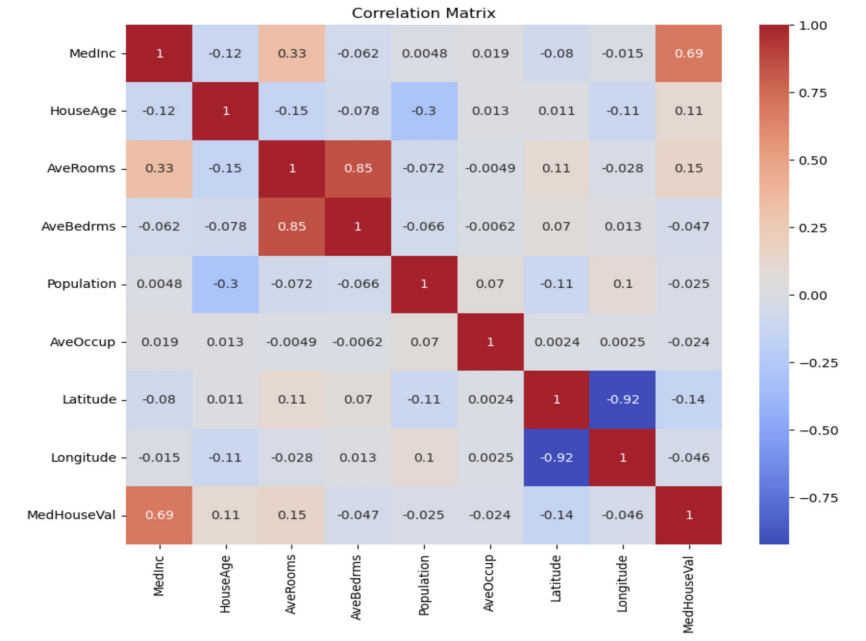

Figure 1: Correlation Heatmap of California Housing Dataset

3. MODEL BUILDING

The California Housing Dataset was divided into:

- Features (X)
- Target variable (y)

The dataset was split into training and testing sets using an 80:20 ratio.

A Linear Regression model from Scikit-learn was trained using the training data. After training, predictions were generated on the testing dataset.

```
# Train Linear Regression Model
model = LinearRegression()

# Train model
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Evaluation
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
```

Figure 2: Linear Regression Model Training and Evaluation Code

The following Python libraries were used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

4. EVALUATION METRICS

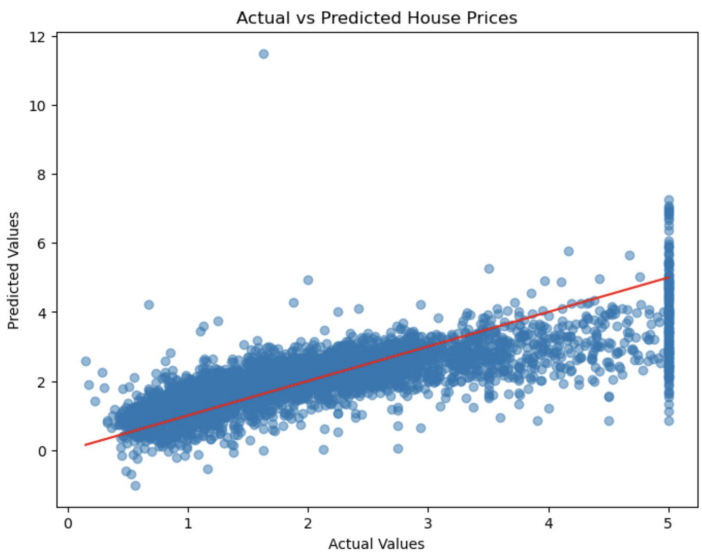

Figure 3: Actual vs Predicted House Prices.

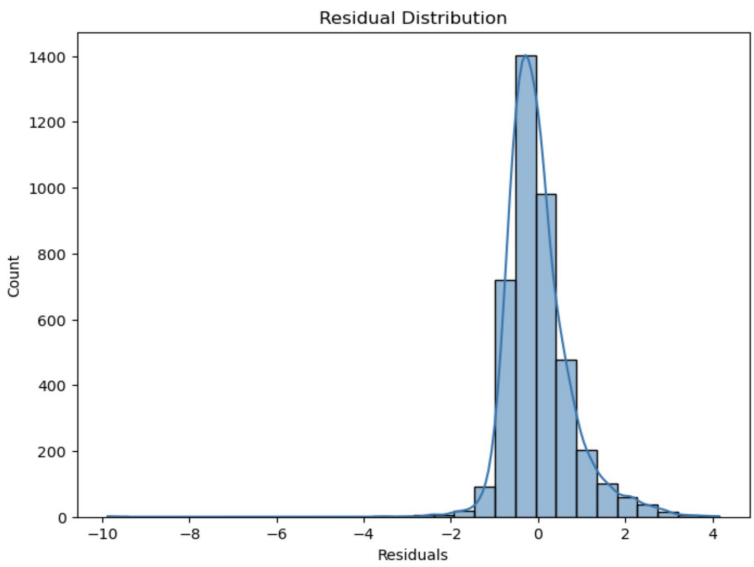

Figure 4: Residual Distribution Plot

The model performance was evaluated using the following regression metrics:

Metric	Value
Mean Absolute Error (MAE)	0.5332
Root Mean Squared Error (RMSE)	0.7456
R^2 Score	0.5758

12. Output

MAE: 0.5332001304956576

RMSE: 0.7455813830127765

R2 Score: 0.5757877060324507

Figure 5: Output of Evaluation Metrics

Interpretation

- MAE indicates moderate prediction error.
- RMSE shows that some predictions have larger deviations.
- The R^2 Score indicates that the model explains approximately 57% of the variance in house prices.

The model performed reasonably well for a basic regression algorithm.

5. IMPROVEMENT IDEAS

The performance of the model can be improved using:

- Feature scaling
- Outlier removal
- Feature engineering
- Hyperparameter tuning

Advanced machine learning algorithms such as:

- Random Forest Regressor
- Decision Tree Regressor
- Gradient Boosting
- XGBoost

can also be used to improve prediction accuracy.

6. CONCLUSION

This project successfully demonstrated the implementation of a Linear Regression model for house price prediction using the California Housing Dataset.

The workflow included:

- Data loading
- Exploratory Data Analysis
- Data preprocessing
- Model training
- Prediction
- Evaluation
- Visualization

The results showed that Linear Regression can effectively predict house prices with moderate accuracy. The project also provided practical understanding of regression analysis and machine learning workflows using Python and Scikit-learn.